

Individual_Assignment_5

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1. Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(NatParksPalettes)

# reading .xlsx files
library(readxl)

# wrap axis labels
library(scales)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # powerful argument in read_xlsx to show the missing variables as numbers
  na = c("", "n/a", "N/A", "over", "173+"))
```

Class Code

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new object from taxon list
taxon_list_clean <- taxon_list |>
  # converted tacon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

```
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
```

```

unite("date_site", date, site, remove = TRUE) |>
# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

```

```

# creating new object
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance") |>
# group by date, site, taxon

```

```

group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# created new column to join with data frames
unite("date_site", date, site, remove = FALSE) |>
# join with water_quality_clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxon_list_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

```

# base layer
salinity_plot <- ggplot(data = aq_ins_clean,
  # x-axis: full site name
  # y-axis: median salinity
  mapping = aes(x = site_full,
                y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points representing median salinity
stat_summary(geom = "point",

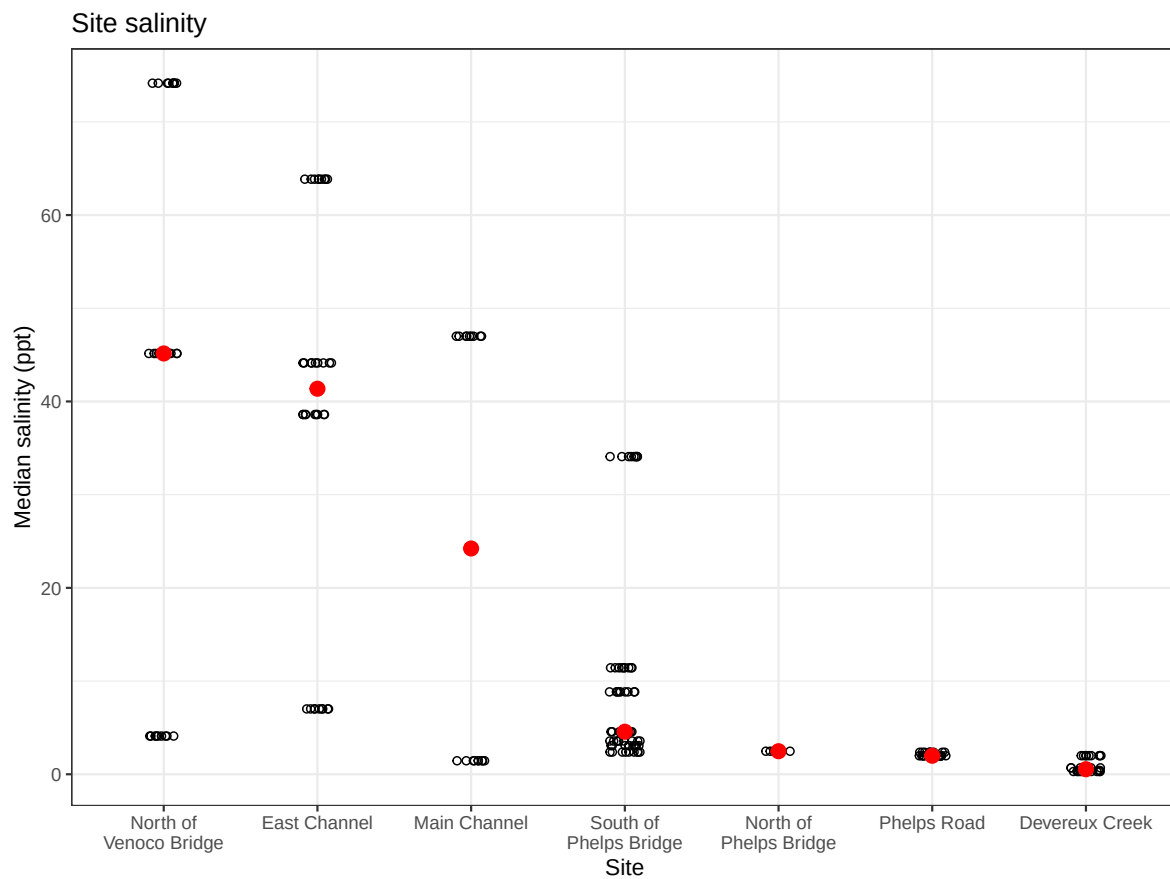
```

```

        fun = median,
        color = "red",
        size = 3) +
# wrapping x-axis labels
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes, adding title
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# using different theme from default
theme_bw()

# displaying plot
salinity_plot

```



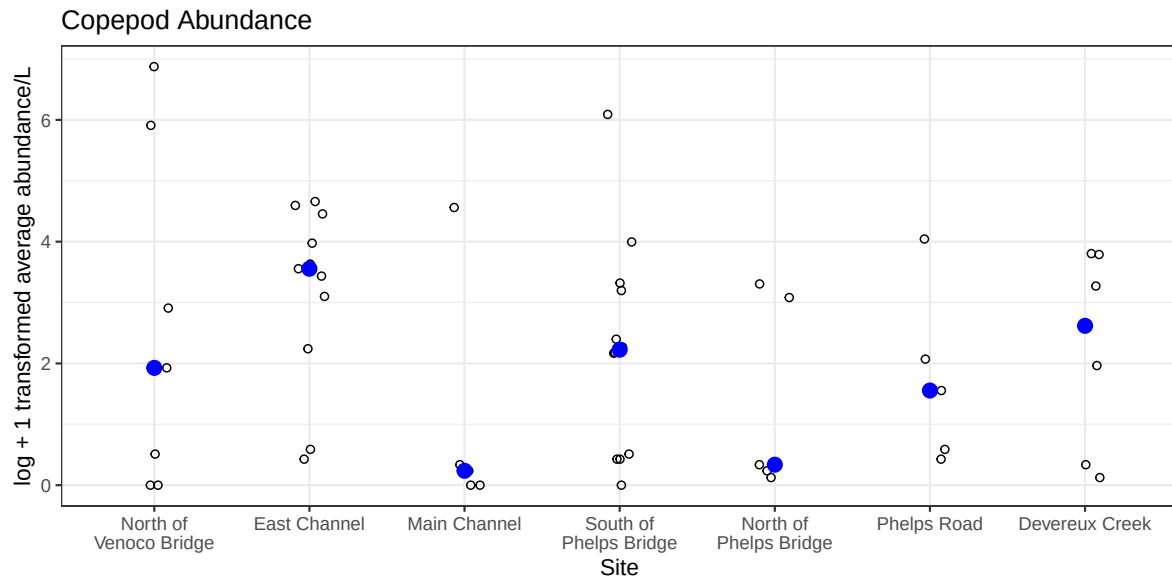
```

copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

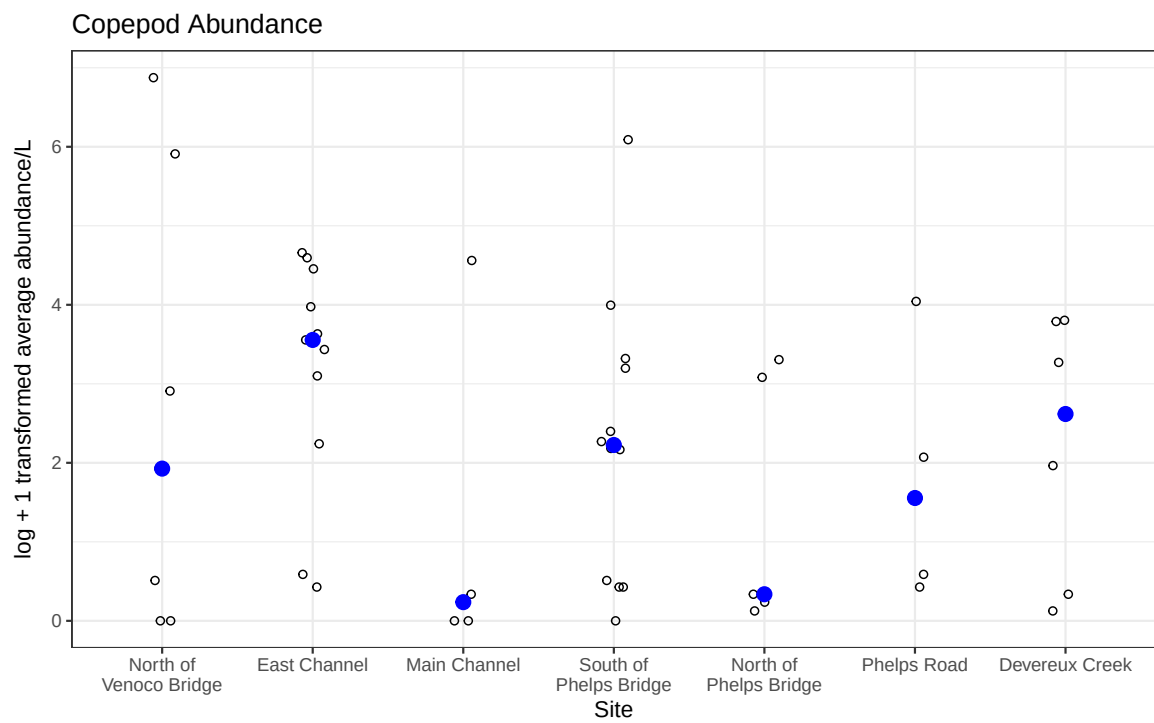
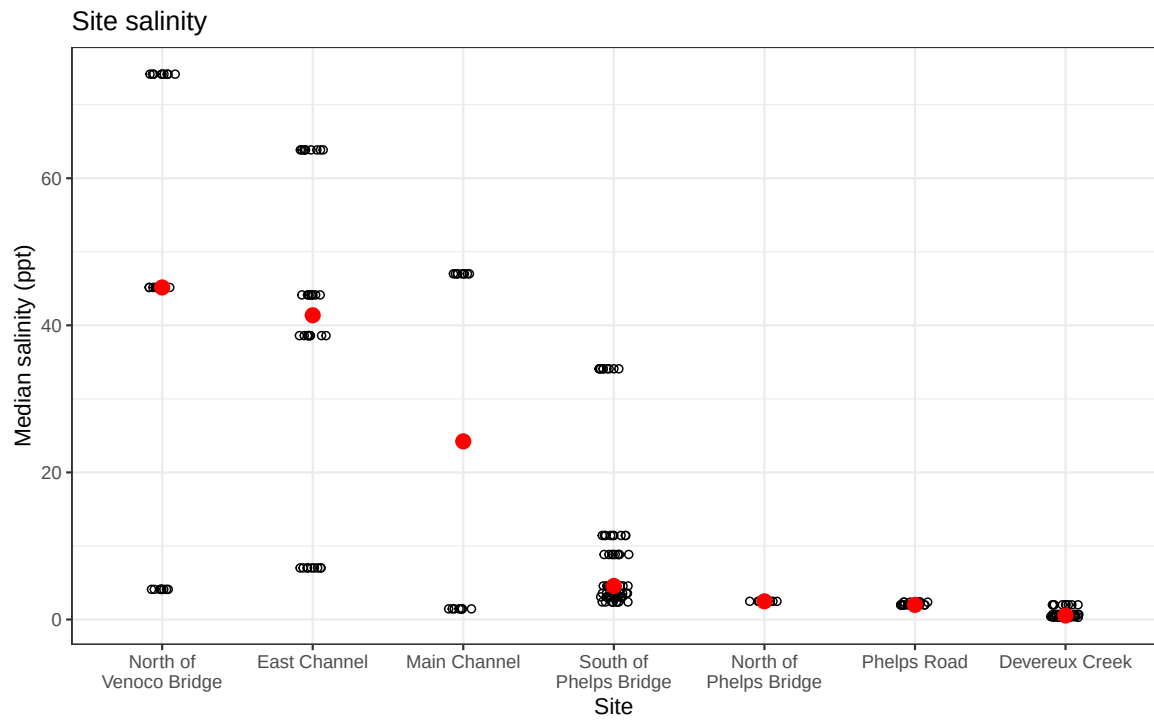
# base layer, creating oobject to store plot
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations, jitter to represent each
  # ↪ sample
  geom_jitter(height = 0,
    shape = 21,
    width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "median",
    size = 3,
    color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
    y = "log + 1 transformed average abundance/L",
    title = "Copepod Abundance") +
  # using different theme
  theme_bw()

# displaying object
copepod_plot

```



`salinity_plot` / `copepod_plot`



Problems

1. Visualizing differences across sites in major taxon abundance

In class, we visualized the differences across sites in median (log transformed) copepod abundance.

In this section, you will create another figure to add to the multipanel plot of salinity and copepod abundance to show ostracod abundance.

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# creating new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filtered to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

c. Code your figure

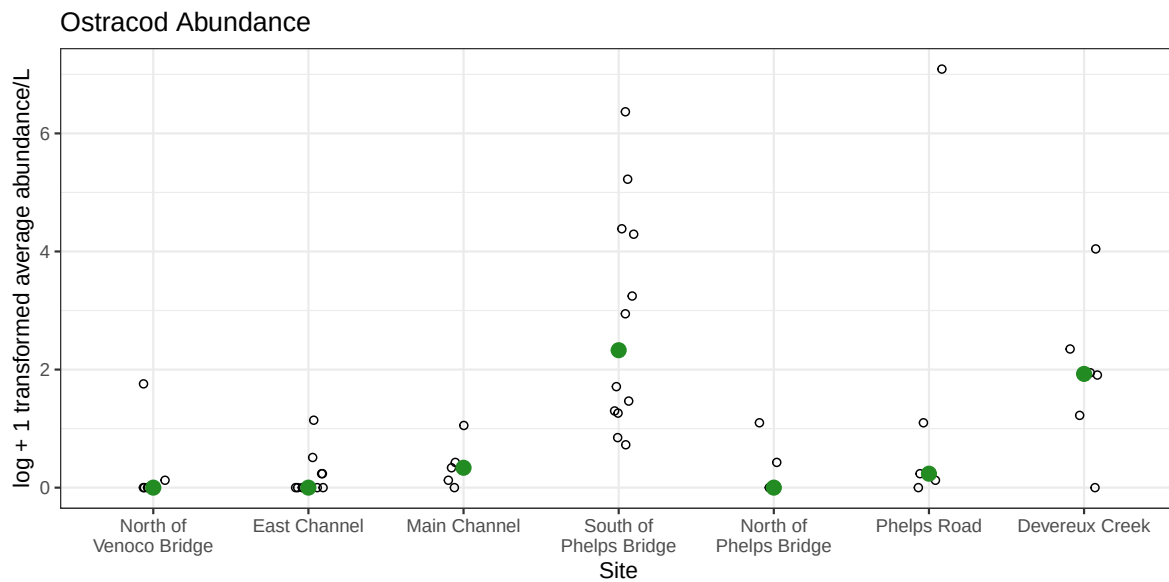
```
# base layer, creating oobject to store plot
ostracod_plot <- ggplot(data = ostracod,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent observations, jitter to represent each
  # sample
  geom_jitter(height = 0,
    shape = 21,
```

```

        width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
             fun = "median",
             size = 3,
             color = "forestgreen") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracod Abundance") +
# using different theme
theme_bw()

# displaying object
ostracod_plot

```

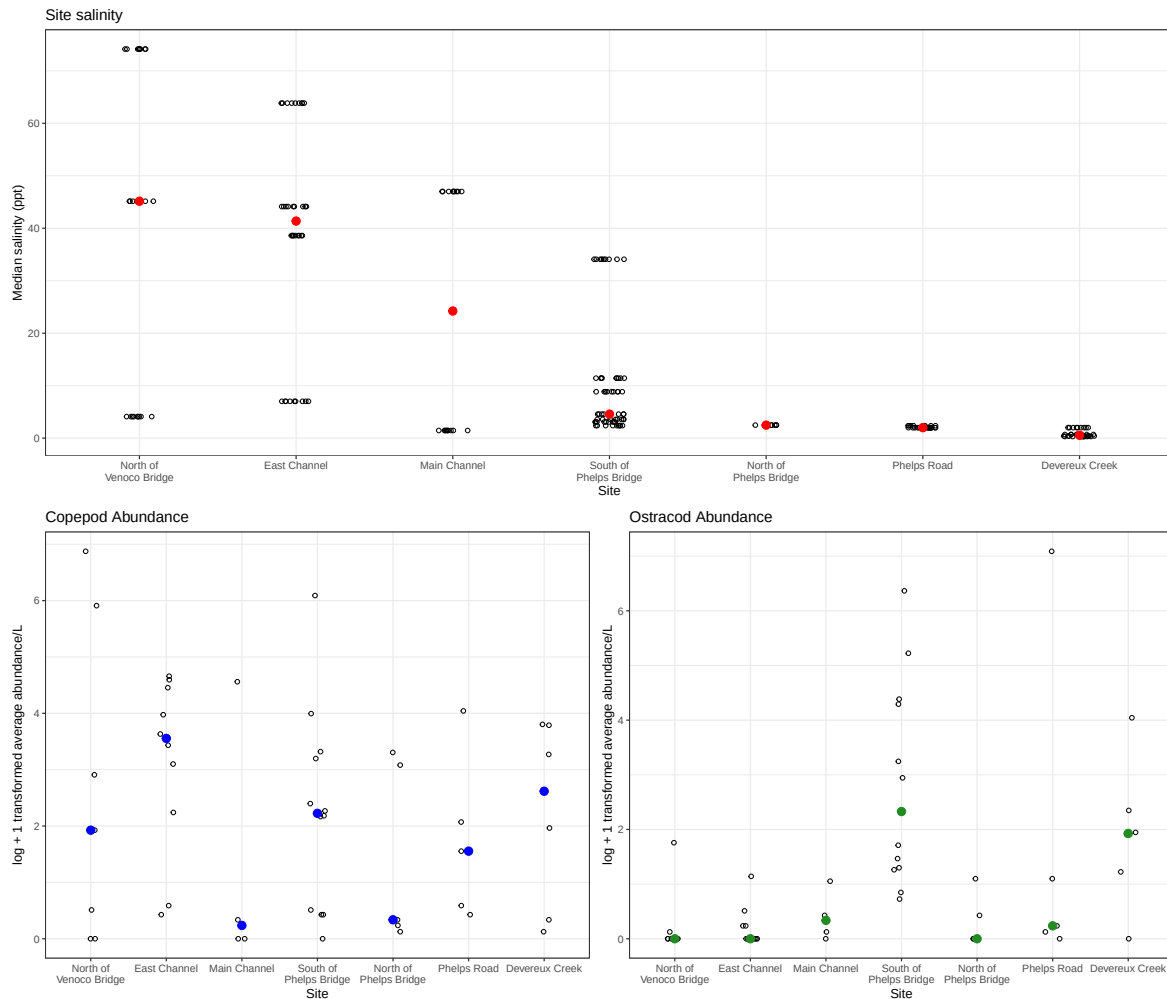


d. Create a multipanel figure

```

# organizing using patchwork arguments
salinity_plot / ( copepod_plot | ostracod_plot )

```



e. Interpret your figure

There is less ostracod than copepod in all sites except for the main channel and South of Phelps bridge where the medians are almost the same (ostracod slightly higher) especially in the South of Phelps bridge since point distributions are much higher. There does not seem to be any correlation between salinity and the abundance of copepods per site due to the randomness of abundance per site.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: ave_lit
- color: taxon
- geometry: geom_point
- facets: taxon

b. Wrangle your data

```
# creating new object from aq_ins_clean
abundance_clean <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # changed taxon names to title case
  mutate(taxon = to_any_case(taxon, case = "title")) |>
  # reorganizing taxons from most to least abundant
  mutate(taxon = fct_reorder(taxon, log_ave_lit, .fun = max, .desc = TRUE))

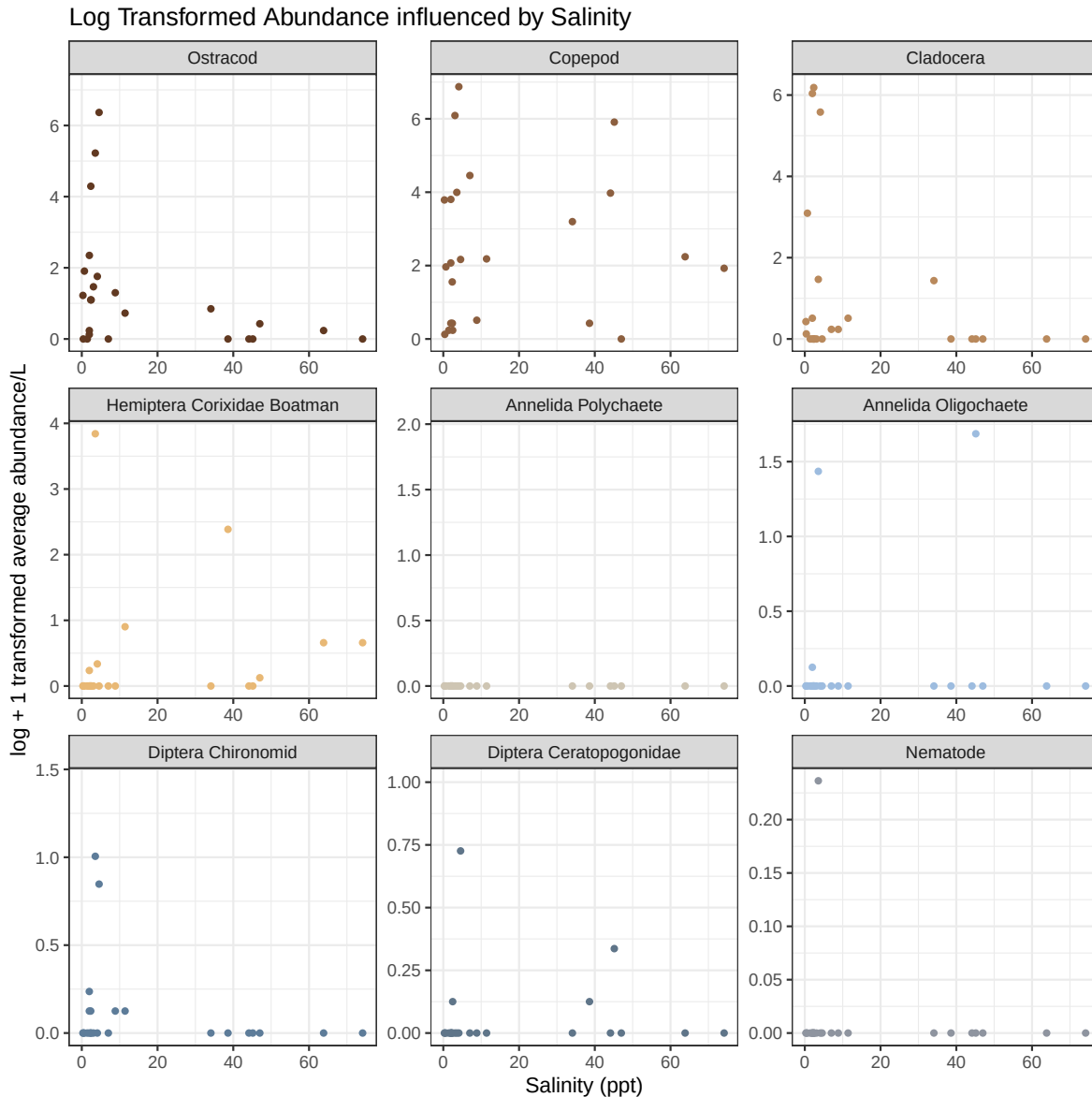
# sample of 10 rows of the cleaned dataset
slice_sample(abundance_clean, n = 10)
```

```
# A tibble: 10 x 24
  date_site      date            site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <fct>  <dbl>  <dbl>  <dbl>  <dbl>
1 2023-05-04_NMC 2023-05-04 00:00:00 NMC    Dipt~  0.267  8.32   NA    10.3
2 2022-05-12_NPB 2022-05-12 00:00:00 NPB    Nema~  0      7.97  11.4   8.17
3 2022-05-12_NPB 2022-05-12 00:00:00 NPB    Dipt~  0.133  7.97  11.4   8.17
4 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1   Dipt~  0      NA     NA     NA
5 2023-05-04_NEC 2023-05-04 00:00:00 NEC    Anne~  0      8.05  NA     8.55
6 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1   Clad~  0      NA     NA     NA
7 2022-01-28_NDC 2022-01-28 00:00:00 NDC    Hemi~  0      7.65  2.00   9.55
8 2023-09-12_NVBR 2023-09-12 00:00:00 NVBR   Dipt~  0.133  9.14  NA     20.3
9 2022-01-21_NEC 2022-01-21 00:00:00 NEC    Dipt~  0      NA     NA     NA
10 2021-08-25_NPB 2021-08-25 00:00:00 NPB    Nema~  0      7.52  4.57   4.5
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer, creating oobject to store plot
abundance_plot <- ggplot(data = abundance_clean,
  # x-axis: full site name
  # y-axis: log transformed average count/L
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = taxon)) +
# first layer: points to represent observations
geom_point(shape = 16) +
# custom colors
scale_color_manual(values = natparks.pals("KingsCanyon", n = 9,
  type = "continuous")) +
# separating by taxon
facet_wrap(~taxon, scales = "free") +
# relabelling x- and y-axis, adding title
labs(x = "Salinity (ppt)",
  y = "log + 1 transformed average abundance/L",
  title = "Log Transformed Abundance influenced by Salinity") +
# using different theme
theme_bw() +
# getting rid of legend
theme(legend.position = "none")

# displaying object
abundance_plot
```



d. Interpret your figure

Across all taxa there is a relatively higher abundance near 0 ppt salinity, suggesting these taxa prefer a lower salinity level to live in. Ostracods, copepods, and cladocera all have the most apparent trend in low salinity preference as the log transformed abundance has multiple observations from 2, all the way to 6. Annelida Oligochaete, Copepod, Hemiptera Corixidae, and Diptera Ceratopogonidae all have a bimodal distribution with peaks or high observations

around 40 ppt. Finally, all taxa show exceptionally low abundance past 60 ppt except for Copepod and Hemiptera Corixidae which show some abundance and tolerance to 60 ppt.